

AI Asset Register Glossary

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Artificial Intelligence (AI)

Description:

Artificial Intelligence (AI) refers to software systems that can perform tasks that normally require human intelligence, such as understanding language, recognizing patterns, making decisions, solving problems, or generating content.

Example:

ChatGPT answering questions, translating languages, summarizing documents, and writing code.

AI System

Description:

An AI System is the complete solution that includes the AI model, datasets, APIs, user interface, infrastructure, monitoring, security controls, and governance processes working together to deliver AI capabilities.

Example:

ChatGPT consists of the GPT model, conversation interface, APIs, authentication, safety guardrails, monitoring, and cloud infrastructure.

Machine Learning (ML)

Description:

Machine Learning (ML) is a subset of AI where algorithms learn patterns from historical data instead of being explicitly programmed with rules.

Example:

A fraud detection system learns from millions of previous transactions to identify fraudulent activity.

Large Language Model (LLM)

Description:

A Large Language Model (LLM) is a deep learning model trained on massive amounts of text to understand and generate human language.

Example:

GPT-5.5 (used by ChatGPT), Claude, Gemini, and Llama.

Foundation Model

Description:

A Foundation Model is a large pre-trained AI model that can be adapted to perform many different tasks without being built from scratch.

Example:

GPT-5.5 can be used for chatbots, summarization, translation, coding assistance, and document analysis.

Model

Description:

A Model is the trained mathematical representation that learns patterns from data and generates predictions or responses.

Example:

GPT-5.5 is the model that generates ChatGPT's responses.

Parameters (Weights)

Description:

Parameters, also called weights, are the learned numerical values inside a model that represent everything it has learned during training. These parameters

determine how the model responds to different inputs.

Example:

When ChatGPT answers a question about cybersecurity, it uses its learned parameters to predict the most appropriate response.

Token

Description:

A Token is the smallest unit of text processed by a language model. A token may represent a word, part of a word, punctuation mark, or special character.

Example:

The sentence "How are you today?" is broken into multiple tokens before ChatGPT processes it.

Tokenizer

Description:

A Tokenizer converts human-readable text into tokens before sending them to the AI model and converts generated tokens back into readable text.

Example:

Every prompt entered into ChatGPT is tokenized before the model begins generating a response.

Context Window

Description:

The Context Window is the maximum amount of information (measured in tokens) that an AI model can process and remember during a conversation.

Example:

ChatGPT remembers earlier messages in a conversation because they are included within its context window.

Prompt

Description:

A Prompt is the instruction, question, or input provided to an AI model to generate a response.

Example:

"Summarize this financial report."

System Prompt

Description:

A System Prompt is a hidden instruction that defines the AI model's behaviour, personality, restrictions, and operating rules before processing user prompts.

Example:

ChatGPT receives internal instructions to be helpful, truthful, and avoid generating harmful content.

Temperature

Description:

Temperature is a model configuration setting that controls how random or creative the generated response will be. Lower values produce more consistent responses, while higher values generate more diverse and creative outputs.

Example:

A bank's chatbot may use a temperature of 0.2 for consistent responses, while a marketing assistant may use 0.9 to generate creative content.

Embedding

Description:

An Embedding is a numerical representation of text, images, or other data that captures its semantic meaning, allowing AI systems to compare information based on similarity rather than exact words.

Example:

A customer support article is converted into an embedding before being searched by ChatGPT.

Vector

Description:

A Vector is the mathematical representation of an embedding consisting of hundreds or thousands of numerical values that describe the meaning of data.

Example:

The sentence "Reset my password" is converted into a vector before being stored in a vector database.

Vector Database

Description:

A Vector Database stores embeddings and performs similarity searches to retrieve the most relevant information for AI applications.

Example:

An enterprise chatbot stores company policies as vectors in Pinecone or pgvector and retrieves them when employees ask questions.

Similarity Search

Description:

Similarity Search compares vectors to find documents or information that are mathematically closest in meaning to the user's query.

Example:

An employee asks about annual leave, and the chatbot retrieves the HR policy even though the exact words do not match.

Retrieval-Augmented Generation (RAG)

Description:

Retrieval-Augmented Generation (RAG) is an AI architecture where the model retrieves relevant information from external knowledge sources before generating a response.

Example:

ChatGPT searches an organization's policy documents before answering an employee's question.

Feature

Description:

A Feature is an individual piece of information used by a machine learning model to make predictions.

Example:

Income, credit score, employment history, and existing loans are features used by a loan approval model.

Feature Store

Description:

A Feature Store is a centralized repository where validated machine learning features are stored and shared across multiple AI models.

Example:

Both the fraud detection model and credit scoring model use customer transaction history stored in the same feature store.

Model Registry

Description:

A Model Registry is a centralized repository used to store, version, approve, and manage trained AI models throughout their lifecycle.

Example:

Only the approved production version of a fraud detection model is deployed from the model registry.

Inference

Description:

Inference is the process of using a trained AI model to generate predictions or responses from new input data.

Example:

Every response generated by ChatGPT is produced through inference.

Hallucination

Description:

A Hallucination occurs when an AI model generates information that appears correct but is factually inaccurate, fabricated, or unsupported.

Example:

ChatGPT creates a legal case citation that does not actually exist.

AI Agent

Description:

An AI Agent is an AI system that can reason, plan, make decisions, and interact with external tools or systems to accomplish a task with limited human intervention.

Example:

An AI travel assistant books flights, reserves hotels, sends confirmation emails, and updates a calendar automatically.

AI Asset Register

Description:

An AI Asset Register is a centralized inventory of all AI-related assets within an organization, including models, datasets, prompts, feature stores, vector databases, model registries, APIs, and supporting infrastructure. It enables governance, risk management, compliance, and security oversight throughout the AI lifecycle.

Example:

A bank's AI Asset Register includes:

- GPT-5.5 Customer Support Model
- Fraud Detection Model
- Training Datasets
- Prompt Templates
- Feature Store
- Vector Database
- Model Registry
- AI APIs
- Monitoring Dashboard